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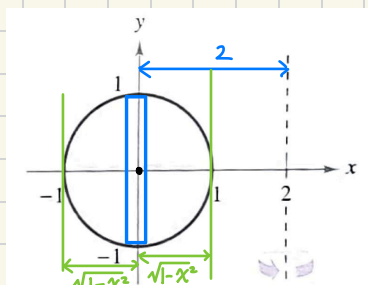
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無論 Disk Method / Shell Method

樣本長方形是鉛直: 對 x 積 (dx) / 是水平: 對 y 積 (dy)

Disk Method: 垂直旋轉軸 / Shell Method: 平行旋轉軸

甜甜圈積分



Shell Method

$$x^2 + y^2 = 1, \quad y^2 = 1 - x^2$$

$$y = \pm \sqrt{1 - x^2}$$

$$p = 2 - x, \quad h(x) = 2\sqrt{1 - x^2}$$

$$p(x)h(x) = (2 - x)(2\sqrt{1 - x^2})$$

$$= 4\sqrt{1 - x^2} - 2x\sqrt{1 - x^2}$$

$$V = 2\pi \int_{-1}^1 (4\sqrt{1 - x^2} - 2x\sqrt{1 - x^2}) dx$$

$$= 8\pi \int_{-1}^1 \sqrt{1 - x^2} dx - 4\pi \int_{-1}^1 x\sqrt{1 - x^2} dx$$

$$= 8\pi \times \frac{\pi}{2} - 0 = 4\pi^2$$

三角代換

Sin 型

$$\sqrt{a^2 - u^2}$$

$$u = a \sin \theta$$

$$\sqrt{a^2 - u^2} = a \cos \theta$$

tan 型

$$\sqrt{a^2 + u^2}$$

$$u = a \tan \theta$$

$$\sqrt{a^2 + u^2} = a \sec \theta$$

sec 型

$$\sqrt{u^2 - a^2}$$

$$u = a \sec \theta$$

$$\sqrt{u^2 - a^2} = \pm a \tan \theta$$

Disk Method

$$R(x)^2 = (2 + \sqrt{1 - x^2})^2 = 4 + 4\sqrt{1 - x^2} + (1 - x^2)$$

$$= 5 + 4\sqrt{1 - x^2} - x^2$$

$$r(x) = (2 - \sqrt{1 - x^2})^2 = 4 - 4\sqrt{1 - x^2} + (1 - x^2)$$

$$= 5 - 4\sqrt{1 - x^2} - x^2$$

$$(-x^2 + 4\sqrt{1 - x^2} + 5) - (-x^2 - 4\sqrt{1 - x^2} + 5)$$

$$= 8\sqrt{1 - x^2}$$

$$V = 8\pi \int_{-1}^1 \sqrt{1 - x^2} dx = 8\pi \times \frac{\pi}{2} = 4\pi^2$$

公式推导:

$$1. \int \sqrt{a^2 - u^2} du = \frac{1}{2} \left(a^2 \arcsin \frac{u}{a} + u \sqrt{a^2 - u^2} \right) + C$$

$$u = a \sin \theta, \quad du = a \cos \theta d\theta, \quad \sin \theta = \frac{u}{a}, \quad \theta = \arcsin \frac{u}{a}$$

$$\sqrt{a^2 - u^2} = a \cos \theta$$

$$\int a \cos \theta a \cos \theta d\theta = a^2 \int \cos^2 \theta d\theta = a^2 \int \frac{1 + \cos 2\theta}{2} d\theta$$

$$= \frac{1}{2} a^2 \int (1 + \cos 2\theta) d\theta \Rightarrow \frac{1}{2} a^2 \int 1 d\theta + \frac{1}{2} a^2 \int \cos 2\theta d\theta$$

$$= \frac{1}{2} a^2 \left(\theta + \frac{1}{2} \sin 2\theta \right)$$

$$= \frac{1}{2} a^2 \left(\arcsin \frac{u}{a} + \frac{1}{2} \times 2 \sin \theta \cos \theta \right) = \frac{1}{2} a^2 \left(\arcsin \frac{u}{a} + \frac{\sqrt{a^2 - u^2}}{a} \times \frac{u}{a} \right)$$

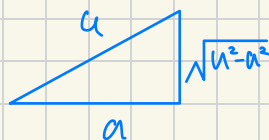
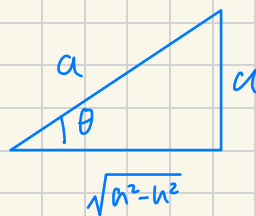
$$= \frac{1}{2} \left(a^2 \arcsin \frac{u}{a} + u \sqrt{a^2 - u^2} \right)$$

$$2. \int \sqrt{u^2 - a^2} du = \frac{1}{2} \left(u \sqrt{u^2 - a^2} - a^2 \ln |u + \sqrt{u^2 - a^2}| \right) + C$$

$$u = a \sec \theta, \quad \sqrt{u^2 - a^2} = \pm a \tan \theta$$

$$du = a \sec \theta \tan \theta d\theta$$

$$\int a \tan \theta a \sec \theta \tan \theta d\theta = \int a^2 \tan^2 \theta \sec \theta d\theta$$



$$S^2 - 1 = t^2$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$3. \int \sqrt{u^2+a^2} du = \frac{1}{2}(u\sqrt{u^2+a^2} - a^2 \ln|u+\sqrt{u^2+a^2}|) + C$$

班級: 資工一 B

科目: 微積分

姓名: _____

學號: _____

1. 求積分 (每小題 8%)

(a) $\int_0^3 \frac{x^3}{\sqrt{x^2+9}} dx$

三角代換

(b) $\int \frac{8x}{x^3 + x^2 - x - 1} dx$

部分分式

(c) $\int x \sec^2 x dx$

I.B.P

(d) $\int_0^1 x \ln x dx$

瑕積分

(e) $\int_0^\infty \frac{e^x}{1+e^x} dx$

$u = 1 + e^x$

$du = e^x dx$

(f) $\int \sqrt{e^t - 2} dt$

$u = \sqrt{e^t - 2}$

$u^2 = e^t - 2$

$e^t = u^2 + 2$

$du = \frac{1}{2} \times (e^t - 2)^{-\frac{1}{2}} \times e^t dt$

(g) $\int_{100}^\infty \frac{dx}{x^{0.998}}$

skip!

(h) $\int_0^{\frac{\pi}{4}} 5 \tan^3 x dx$

一個換 $\sec^2 \theta - 1$

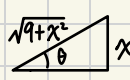
剩下 let $u = \tan \theta$

(i) $\int e^x (e^x + 1)^2 dx$

$u = e^x + 1$

(j) $\int \frac{e^{2x}}{4 + e^{4x}} dx$

$u = e^{2x}$

$$(a) \int_0^3 \frac{x^3}{\sqrt{x^2+9}} dx$$


let $\sqrt{9+x^2} = 3\sec\theta$, $\sec\theta = \frac{\sqrt{9+x^2}}{3}$
 $x = 3\tan\theta$, $dx = 3\sec^2\theta d\theta$

$$\int \frac{27\tan^3\theta}{3\sec\theta} \times 3\sec^2\theta d\theta$$

$$= 27 \int \tan^3\theta \sec\theta d\theta$$

$$= 27 \int (\sec^2\theta - 1) \tan\theta \sec\theta d\theta$$

let $u = \sec\theta$, $du = \tan\theta \sec\theta d\theta$

$$= 27 \int (u^2 - 1) du = 27 \left(\frac{1}{3} u^3 - u \right)$$

$$= 9u^3 - 27u = 9x \frac{(9+x^2)^{\frac{3}{2}}}{3} - 27x \frac{(9+x^2)^{\frac{1}{2}}}{3}$$

$$= \frac{(9+x^2)^{\frac{3}{2}}}{3} - 9(9+x^2)^{\frac{1}{2}}$$

$$\Rightarrow \left(\frac{1}{3} (9+x^2)^{\frac{3}{2}} - 9(9+x^2)^{\frac{1}{2}} \right) \Big|_0^3$$

$$= \left(\frac{1}{3} 18\sqrt{18} - 9\sqrt{18} \right) - \left(\frac{1}{3} 9\sqrt{9} - 9\sqrt{9} \right)$$

$$= -3\sqrt{18} + 6\sqrt{9} = -9\sqrt{2} + 18 = 18 - 9\sqrt{2}$$

$$(c) \int \frac{x \sec^2 x}{u} dx$$

let $u = x$, $dv = \sec^2 x dx$

u	dv
$+x$	$\sec^2 x$
-1	$\tan x$
$+0$	$-\ln \cos x $

$$\Rightarrow x \tan x + \ln|\cos x| + C$$

$$(d) \int_0^1 x \ln x dx$$

$$\lim_{b \rightarrow 0^+} \int_b^1 \frac{\ln x}{u} dv = \lim_{b \rightarrow 0^+} \left(\frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 \right) \Big|_b^1 = \lim_{b \rightarrow 0^+} \left[\left(\frac{1}{2} \ln 1 - \frac{1}{4} \right) - \left(\frac{1}{2} b^2 \ln b - \frac{1}{4} b^2 \right) \right]$$

$$u = \ln x, dv = x dx$$

$$du = \frac{1}{x} dx, v = \frac{1}{2} x^2 \Rightarrow \frac{\ln x}{2} x^2 - \int \frac{1}{2} x \times \frac{1}{x} dx$$

$$= \lim_{b \rightarrow 0^+} \left[\frac{1}{2} \ln b - \frac{1}{4} (b^2 - 1) \right] = -\frac{1}{4}$$

$$= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2$$

$$(b) \int \frac{8x}{x^3+x^2-x-1} dx$$

$$(x^3+x^2-x-1) = (x-1)(x^2+2x+1)$$

$$= (x-1)(x+1)^2$$

$$\frac{8x}{x^3+x^2-x-1} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

$$8x = A(x+1)^2 + B(x-1)(x+1) + C(x-1)$$

$$x=-1, -8 = -2C, C=4$$

$$x=1, 8 = 4A, A=2$$

$$x=2, 16 = 18 + 3B + 4, -6 = 3B, B=-2$$

$$\int \frac{2}{x-1} + \frac{-2}{x+1} + \frac{4}{(x+1)^2} dx$$

$$= 2 \ln|x-1| - 2 \ln|x+1| - \frac{4}{x+1} + C$$

$$(e) \int_0^\infty \frac{e^x}{1+e^x} dx$$

$$u = 1+e^x, du = e^x dx$$

$$\int \frac{1}{u} du = \ln|u| = \ln|1+e^x|$$

$$\lim_{a \rightarrow \infty} (\ln|1+e^x|) \Big|_0^a = \lim_{a \rightarrow \infty} (\ln|1+e^a| - \ln|2|)$$

$\xrightarrow{e^\infty = \infty}$

$$= \lim_{a \rightarrow \infty} (\infty - \ln 2) = \infty$$

Ans: ∞ (發散)

$$(f) \int \sqrt{e^t - 2} dt$$

$$u = \sqrt{e^t - 2}, u^2 = e^t - 2, e^t = u^2 + 2$$

$$du = \frac{1}{2} e^t (e^t - 2)^{-\frac{1}{2}} dt$$

$$= \frac{e^t}{2\sqrt{e^t - 2}} dt$$

$$dt = \frac{2\sqrt{e^t - 2}}{e^t} du$$

$$\int u \times \frac{2u}{u^2 + 2} du = 2 \int \frac{u^2}{u^2 + 2} du$$

$$= 2 \left(\int \frac{u^2 + 2 - 2}{u^2 + 2} du - \int \frac{2}{u^2 + 2} du \right)$$

$$= 2 \left(u - 2 \int \frac{du}{u^2 + 2} \right) = 2 \left(u - \frac{2}{\sqrt{2}} \arctan \frac{u}{\sqrt{2}} \right)$$

$$= 2u - 2\sqrt{2} \arctan \frac{u}{\sqrt{2}} = 2\sqrt{e^t - 2} - 2\sqrt{2} \arctan \frac{\sqrt{e^t - 2}}{\sqrt{2}} + C$$

$$(g) \int e^x (e^x + 1)^2 dx$$

$$\text{let } u = e^x + 1$$

$$du = e^x dx$$

$$\int u^2 du = \frac{1}{3} u^3 + C$$

$$= \frac{1}{3} (e^x + 1)^3 + C$$

$$(h) \int_0^{\frac{\pi}{4}} 5 \tan^3 x dx$$

$$= \int_0^{\frac{\pi}{4}} 5 (\sec^2 x - 1) \tan x dx$$

$$= 5 \int_0^{\frac{\pi}{4}} \tan x \sec^2 x - \tan x dx$$

$$\left[\int \tan x \sec^2 x dx, u = \tan x, du = \sec^2 x \right]$$

$$\Rightarrow \int u du = \frac{1}{2} u^2 = \frac{1}{2} \tan^2 x$$

$$\Rightarrow 5 \left(\frac{1}{2} \tan^2 x + \ln |\cos x| \right) \Big|_0^{\frac{\pi}{4}}$$

$$= 5 \left(\left(\frac{1}{2} \times 1^2 + \ln \left| \frac{\sqrt{2}}{2} \right| \right) - \left(0 + \ln |1| \right) \right)$$

$$= 5 \times \left(\frac{1}{2} + \ln \left| \frac{\sqrt{2}}{2} \right| \right) = \frac{5}{2} + 5 \ln \left| \frac{\sqrt{2}}{2} \right|$$

$$(i) \int \frac{e^{2x}}{4 + e^{4x}} dx$$

$$u = e^{2x}, du = 2e^{2x} dx, \frac{du}{2} = e^{2x} dx$$

$$\Rightarrow \int \frac{1}{4 + u^2} \times \frac{du}{2} = \frac{1}{2} \int \frac{1}{4 + u^2} du = \frac{1}{2} \int \frac{1}{2^2 + u^2} du$$

$$\Rightarrow \frac{1}{2} \times \left(\frac{1}{2} \arctan \frac{u}{2} \right) = \frac{1}{4} \arctan \frac{u}{2} + C$$

$$\Rightarrow \frac{1}{4} \arctan \frac{e^{2x}}{2} + C$$

班級: 資工一 B

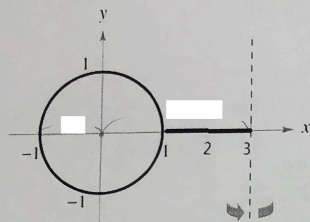
科目: 微積分

姓名: _____

學號: _____

1. 用 disk 或 shell 方法求下圖甜甜圈體積: 即將 $x^2 + y^2 \leq 1$ 區域繞轉軸 $x=2$ 旋轉一圈所成體積.

(10%)



10

2. 求下列曲線與 x 軸所圍區域, 繞 x 軸旋轉的側表面積: (10%)

$$y = \frac{x^3}{6} + \frac{1}{2x}, \quad 1 \leq x \leq 2$$

3. 求曲線長: $x = \frac{1}{3}(y^2 + 2)^{3/2}, 0 \leq y \leq 3$. (9%)

4. 求函數 $f(x) = \ln x$ 在 $C=1$ 的四次泰勒多項式, 並用它估算 $\ln 0.9$, 準確到小數點後第三位, 不能用計算器. (12%)

1.

$$x^2 + y^2 = 1$$

$$y^2 = x^2 - 1, \quad y = \pm \sqrt{x^2 - 1}$$

$$p = 3 - x, \quad h(x) = 2\sqrt{x^2 - 1}$$

$$p(x)h(x) = (3-x)(2\sqrt{x^2-1}) \\ = 6\sqrt{x^2-1} - 2x\sqrt{x^2-1}$$

$$V = 2\pi \int_{-1}^1 6\sqrt{x^2-1} - 2x\sqrt{x^2-1} \, dx$$

$$= 12\pi \int_{-1}^1 \sqrt{x^2-1} \, dx - 4\pi \int_{-1}^1 x\sqrt{x^2-1} \, dx$$

$$= \overset{6}{12}\pi \times \frac{\pi}{2} = 6\pi^2$$

1. 求積分: (a) $\int_0^{\frac{\pi}{4}} 5 \tan^3 x \, dx$

(b) $\int_0^1 \ln(4+x^2) dx$

(c) $\int_0^2 \frac{1}{x^2 - 2x + 2} dx$

(d) 求 $G'(x) = ?$, 其中 $G(x) = \int_{2023}^{x^3-1} \cos(t^2+t-2) \, dt$.

(e) $\int_0^2 \frac{x}{\sqrt{1+2x^2}} dx$

(f) $\int \sin^3 x \cos^2 x \, dx$

(g) $\int e^{-3x} \sin 2x \, dx$

(h) $\int e^x \sqrt{1-e^{2x}} \, dx$

(i) $\int e^x (e^x + 1)^2 \, dx$

2. 說明(推導):

$$\int \sqrt{a^2 - u^2} \, du = \frac{1}{2} \left(a^2 \arcsin \frac{u}{a} + u \sqrt{a^2 - u^2} \right) + C$$

$$1. (a) \int_0^{\frac{\pi}{4}} 5 \tan^3 x \, dx$$

$$= 5 \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \tan x \, dx$$

$$= 5 \int_0^{\frac{\pi}{4}} \tan x \sec^2 x - \tan x \, dx$$

$$\int \tan x \sec^2 x \, dx \quad u = \tan x \quad du = \sec^2 x \, dx$$

$$\int u \, du = \frac{1}{2} u^2 + C = \frac{1}{2} \tan^2 x$$

$$= 5 \left(\frac{1}{2} \tan^2 x + \ln |\cos x| \right) \Big|_0^{\frac{\pi}{4}}$$

$$= 5 \left[\left(\frac{1}{2} \times 1^2 - \ln \left| \frac{\sqrt{2}}{2} \right| \right) + (0 - \ln 1) \right]$$

$$= \frac{5}{2} + 5 \ln \left| \frac{\sqrt{2}}{2} \right| \quad \#$$

$$1. (c) \int_0^2 \frac{1}{x^2 - 2x + 2} \, dx$$

$$(x^2 - 2x + 1) + 2 - 1$$

$$= (x-1)^2 + 1$$

$$\int_0^2 \frac{1}{1 + (x-1)^2} \, dx$$

$$= \left(\arctan(x-1) \right) \Big|_0^2$$

$$= (\arctan(1) - 0) = \frac{\pi}{4} \quad \#$$

$$1. (b) \int_0^1 \frac{\ln(4+x^2)}{x^2+4} \, dx \quad u = \ln(4+x^2), \quad du = \frac{2x}{4+x^2} \, dx$$

$$dv = dx, \quad v = x$$

$$x \ln(4+x^2) - \int \frac{2x^2}{x^2+4} \, dx$$

$$= x \ln(4+x^2) - 2 \left(\int \frac{x^2+4}{x^2+4} \, dx - \int \frac{4}{x^2+4} \, dx \right)$$

$$= x \ln(4+x^2) - 2 \left(x - 2 \arctan \frac{x}{2} \right)$$

$$= x \ln(4+x^2) - 2x + 4 \arctan \frac{x}{2}$$

$$= \left(x \ln(4+x^2) - 2x + 4 \arctan \frac{x}{2} \right) \Big|_0^1$$

$$= \left[(1 \times \ln 5) - 2 + 4 \arctan \frac{1}{2} \right] - (0)$$

$$= \ln 5 - 2 + 4 \arctan \frac{1}{2}$$

$$1. (e) \int_0^2 \frac{x}{\sqrt{1+2x^2}} \, dx$$

$$u = 1+2x^2, \quad du = 4x \, dx, \quad \frac{du}{4} = x \, dx$$

$$\frac{1}{4} \int_1^9 u^{\frac{1}{2}} \, du = \frac{1}{4} \left(\frac{2}{3} u^{\frac{3}{2}} \right) \Big|_1^9$$

$$= \frac{1}{4} \left(\frac{18 \sqrt{9}}{3} - \frac{2}{3} \right) = \frac{1}{4} \times \frac{52}{3} = \frac{13}{3} \quad \#$$

$$1. (f) \int \sin^2 x \cos^2 x \, dx$$

$$u = \cos x, \quad du = -\sin x \, dx$$

$$dx = \frac{1}{-\sin x} \, du$$

$$\int \sin^2 x \cos^2 x \times \frac{1}{-\sin x} \, du = - \int \sin x \cos^2 x \, du$$

$$= - \int (1 - \cos^2 x) \cos^2 x \, du = \int (1 - u^2) u^2 \, du = \int u^2 - u^4 \, du$$

$$= - \left(\frac{1}{3} u^3 - \frac{1}{5} u^5 \right) = \frac{1}{5} u^5 - \frac{1}{3} u^3 + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C \quad \#$$

$$(g) \int e^{-3x} \sin 2x dx$$

$$\Rightarrow \int \underbrace{\sin 2x}_u \underbrace{e^{-3x} dx}_{dv}$$

$$u = \sin 2x, \quad du = 2 \cos 2x dx$$

$$dv = e^{-3x} dx, \quad v = -\frac{1}{3} e^{-3x}$$

$$\Rightarrow \sin 2x \cdot -\frac{1}{3} e^{-3x} - \int -\frac{1}{3} e^{-3x} 2 \cos 2x dx$$

$$= -\frac{1}{3} \sin 2x e^{-3x} + \frac{2}{3} \int e^{-3x} \cos 2x dx$$